

```
PS C:\Users\clouduser\divya\mongodb_replication_demo\scripts2> mongosh --port 27018
rs0 [direct: primary] assessDB> show collections
activityLog
comments
followers
posts
users
rs0 [direct: primary] assessDB> []
```

```
mongodb_replication_demo > scripts2 > JS setupreplica.js > ...
```

```
1  rs.initiate({
2    _id: "rs0",
3
4    members: [
5      { _id: 0, host: "localhost:27017"},
6      { _id: 1, host: "localhost:27018" },
7      { _id: 2, host: "localhost:27019" }
8    ]
9  });
10
11
12  print("Replica Set Initialized");
13  |
```

```

1 use("assessDB");
2
3 db.users.insertMany([
4   {
5     username: "alex_jones",
6     email: "alex@example.com",
7     createdAt: new Date(),
8     profile: { bio: "Tech enthusiast & blogger", location: "New York" }
9   },
10  {
11    username: "sam_smith",
12    email: "sam@example.com",
13    createdAt: new Date(),
14    profile: { bio: "Digital artist | Designer", location: "London" }
15  },
16  {
17    username: "maria_v",
18    email: "maria@example.com",
19    createdAt: new Date(),
20    profile: { bio: "Traveler and food lover", location: "Paris" }
21  }
22 ]);
23 print("Data inserted successfully");
24

```

```

1 use("assessDB");
2
3 db.posts.insertMany([
4   {
5     postId: "p991",
6     userId: "u101",
7     caption: "Setting up a 3-node MongoDB replica set today! Super smooth.",
8     hashtags: ["mongodb", "database", "replication"],
9     likesCount: 45,
10  },
11  {
12    postId: "p992",
13    userId: "u102",
14    caption: "Check out my new UI design layout portfolio updates!",
15    hashtags: ["uiux", "design", "portfolio"],
16    likesCount: 112,
17  }
18 ]);
19
20 print("Successfully initialized posts collection!");

```

```
use("assessDB");
db.comments.insertMany([
  {
    commentId: "c_001",
    postId: "p991",
    userId: "u102",
    commentText: "Brilliant setup! Redundancy is key.",
  },
  {
    commentId: "c_002",
    postId: "p991",
    userId: "u101",
    commentText: "Agreed! It works perfectly.",
  }
]);

print("Successfully initialized comments collection!");
```

```
use("assessDB");
db.followers.insertMany([
  {
    userId: "u101",
    followerId: "u102",
    timestamp: new Date()
  }
]);

print("Successfully initialized followers collection!");
```

```

use("assessDB");
db.activityLog.insertMany([
  {
    logId: "log_7701",
    userId: "u101",
    activityType: "Post creation",
    device: "MacBook Pro 16",
    ipAddress: "192.168.1.45",
    location: "San Francisco",
    timestamp: new Date()
  },
  {
    logId: "log_7702",
    userId: "u102",
    activityType: "Comment activity",
    device: "iPhone 15 Pro",
    ipAddress: "172.56.12.88",
    location: "Austin",
    timestamp: new Date()
  }
]);

print("Successfully initialized activity logs collection!");

```

```

s0 [direct: primary] assessDB> db.users.find()
{
  _id: ObjectId('6a150731e7f179db9cab114'),
  username: 'alex_jones',
  email: 'alex@example.com',
  createdAt: ISODate('2026-05-26T02:36:33.574Z'),
  profile: { bio: 'Tech enthusiast & blogger', location: 'New York' }
},
{
  _id: ObjectId('6a150731e7f179db9cab115'),
  username: 'sam_smith',
  email: 'sam@example.com',
  createdAt: ISODate('2026-05-26T02:36:33.574Z'),
  profile: { bio: 'Digital artist | Designer', location: 'London' }
},
{
  _id: ObjectId('6a150731e7f179db9cab116'),
  username: 'maria_v',
  email: 'maria@example.com',
  createdAt: ISODate('2026-05-26T02:36:33.574Z'),
  profile: { bio: 'Traveler and food lover', location: 'Paris' }
}

```

```
rs0 [direct: primary] assessDB> db.posts.find()
[
  {
    _id: ObjectId('6a1508ce6cb70f5accabc114'),
    postId: 'p991',
    userId: 'u101',
    caption: 'Setting up a 3-node MongoDB replica set today! Super smooth.',
    hashtags: [ 'mongodb', 'database', 'replication' ],
    likesCount: 45
  },
  {
    _id: ObjectId('6a1508ce6cb70f5accabc115'),
    postId: 'p992',
    userId: 'u102',
    caption: 'Check out my new UI design layout portfolio updates!',
    hashtags: [ 'uiux', 'design', 'portfolio' ],
    likesCount: 112
  }
]
```

```
rs0 [direct: primary] assessDB> db.comments.find()
[
  {
    _id: ObjectId('6a1509325b1fc91c4babc114'),
    commentId: 'c_001',
    postId: 'p991',
    userId: 'u102',
    commentText: 'Brilliant setup! Redundancy is key.'
  },
  {
    _id: ObjectId('6a1509325b1fc91c4babc115'),
    commentId: 'c_002',
    postId: 'p991',
    userId: 'u101',
    commentText: 'Agreed! It works perfectly.'
  }
]
```

```
rs0 [direct: primary] assessDB> db.followers.find()
[
  {
    _id: ObjectId('6a15098f51dab12058abc114'),
    userId: 'u101',
    followerId: 'u102',
    timestamp: ISODate('2026-05-26T02:46:39.789Z')
  }
]
```



```
rs0 [direct: primary] assessDB> db.activityLog.find()
[
  {
    _id: ObjectId('6a150a042dc799e41eabc114'),
    logId: 'log_7701',
    userId: 'u101',
    activityType: 'Post creation',
    device: 'MacBook Pro 16',
    ipAddress: '192.168.1.45',
    location: 'San Francisco',
    timestamp: ISODate('2026-05-26T02:48:36.114Z')
  },
  {
    _id: ObjectId('6a150a042dc799e41eabc115'),
    logId: 'log_7702',
    userId: 'u102',
    activityType: 'Comment activity',
    device: 'iPhone 15 Pro',
    ipAddress: '172.56.12.88',
    location: 'Austin',
    timestamp: ISODate('2026-05-26T02:48:36.115Z')
  }
]
```

```
use("assessDB");

db.users.updateOne(
  { email: "alex@example.com" },
  { $set: { bio: "Updated Security Architect Bio!" } }
);
print("User bio updated successfully.");
db.posts.updateOne(
  { postId: "p991" },
  { $set: { caption: "This is a newly updated caption for my post!" } }
);
print("Post caption updated successfully.");
```

```
rs0 [direct: primary] assessDB> db.users.find()
[
  {
    _id: ObjectId('6a150731e7f179db9cabc114'),
    username: 'alex_jones',
    email: 'alex@example.com',
    createdAt: ISODate('2026-05-26T02:36:33.574Z'),
    profile: { bio: 'Tech enthusiast & blogger', location: 'New York' },
    bio: 'Updated Security Architect Bio!'
  },
  {

```

```
rs0 [direct: primary] assessDB> db.posts.find()
[
  {
    _id: ObjectId('6a1508ce6cb70f5accabc114'),
    postId: 'p991',
    userId: 'u101',
    caption: 'This is a newly updated caption for my post!',
    hashtags: [ 'mongodb', 'database', 'replication' ],
    likesCount: 45
  },
  {
    _id: ObjectId('6a1508ce6cb70f5accabc114'),
    postId: 'p991',
    userId: 'u101',
    caption: 'This is a newly updated caption for my post!',
    hashtags: [ 'mongodb', 'database', 'replication' ],
    likesCount: 45
  }
]
```

```
mongodb_replication_demo > scripts2 > JS delData.js > ...
```

```
1 use("assessDB");
2
3 db.posts.deleteOne({ postId: "p992" });
4 print("Post deleted successfully.");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\clouduser\divya\mongodb_replication_demo\scripts2> mongosh --port 27018
with --bind_ip <address> to specify which IP addresses it should serve responses from, or with --bind_ip_all to bind to all interfaces. If this behavior is desired, start the server with --bind_ip 127.0.0.1 to disable this warning

```
rs0 [direct: primary] test> use assessDB
switched to db assessDB
rs0 [direct: primary] assessDB> db.posts.find()
[
  {
    _id: ObjectId('6a1508ce6cb70f5accabc114'),
    postId: 'p991',
    userId: 'u101',
    caption: 'This is a newly updated caption for my post!',
    hashtags: [ 'mongodb', 'database', 'replication' ],
    likesCount: 45
  }
]
```